SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA0509**

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 40%

QUESTIONS		
Q. No	**Question**	**Bloom's Level**
1	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember
2	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand
3	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand
4	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand
5	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember

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Code of Conduct:

I **<u>VENNA MEENAKSHI REDDY - 192471048</u>** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: VENNA MEENAKSHI REDDY

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.

Schema

A schema is the overall logical structure or blueprint of a database. It defines the tables, attributes, data types, relationships, and constraints. A schema is designed once and changes very rarely.

Example

Student Table

Attribute	Data Type
Student_ID	INT
Name	VARCHAR(50)
Department	VARCHAR(30)
Age	INT

The definition of this table (Student_ID INT, Name VARCHAR(50), Department VARCHAR(30), Age INT) is the **schema**.

Instance

An instance is the actual data stored in the database at a particular moment. Unlike the schema, the instance changes whenever records are inserted, updated, or deleted.

Example

Student_ID	Name	Department	Age
101	Rahul	CSE	20
102	Priya	ECE	19

These records represent one database instance.

Difference Between Schema and Instance

Schema	Instance
Blueprint of the database	Actual data stored
Changes rarely	Changes frequently
Defined during database design	Changes during database operations
Specifies structure	Represents current contents

Types of Schema

1. Logical Schema

The logical schema defines the logical structure of the database, including tables, attributes, relationships, and constraints. It does not describe how data is physically stored.

Example

STUDENT(Student_ID, Name, Department)

COURSE(Course_ID, Course_Name)

Relationship:

STUDENT ---- Enrolls ---- COURSE

2. Physical Schema

The physical schema specifies how the data is stored on storage devices, including file organization, indexes, hashing, and storage blocks.

Example

- Student table stored in file Student.dat
- B+ Tree index on Student_ID
- Records stored in 4 KB disk blocks

3. View Schema (External Schema)

The view schema presents only the required portion of the database to specific users, improving security and simplicity.

Example

Faculty View:

Student_ID	Name
101	Rahul
102	Priya

The Age and Department columns are hidden.

Comparison of Schemas

Logical Schema	Physical Schema	View Schema
Defines database structure	Defines storage details	Defines user-specific view
Used by database designers	Used by DBMS	Used by end users
Independent of storage	Depends on storage	Shows only required data
Example: Tables and relationships	**Example:** Indexes, files	**Example:** Student marks view

Conclusion

A schema defines the structure of the database, whereas an instance represents the actual data. The logical schema describes the database design, the physical schema specifies storage details, and the view schema provides customized views for different users, ensuring security and simplicity.

2. Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.

Introduction

A Bank Management System is used to manage customer information, accounts, loans, payments, employees, and branch details. An Entity-Relationship (ER) diagram represents the data requirements of the system by identifying the entities, attributes, and relationships among them.

1. Entities and Attributes

1. CUSTOMER

Primary Key: Customer_ID

Attributes:

- Customer_ID (PK)
- Customer_Name
- Customer_Street
- Customer_City
- Customer_Dependent_Name

2. BRANCH

Primary Key: Branch_Name

Attributes:

- Branch_Name (PK)
- Branch_City
- Assets

3. LOAN

Primary Key: Loan_Number

Attributes:

- Loan_Number (PK)

4. PAYMENT

Primary Key: Payment_Number

Attributes:

- Payment_Number (PK)
- Amount
- Date_of_Payment

5. ACCOUNT

Primary Key: Account_Number

Attributes:

- Account_Number (PK)
- Balance

6. EMPLOYEE

Primary Key: Employee_ID

Attributes:

- Employee_ID (PK)
- Employee_Name
- Employee_Experience

7. SAVINGS ACCOUNT *(Subclass of Account)*

Attribute:

- Interest_Rate

8. CHECKING ACCOUNT *(Subclass of Account)*

Attribute:

- Overdraft_Amount

2. Relationships

1. CUSTOMER — Borrower — LOAN

- One customer can borrow one or more loans.
- Each loan is borrowed by one customer.
- Cardinality: 1 : M

2. BRANCH — Loan_Branch — LOAN

- One branch can issue many loans.
- Each loan is issued by one branch.
- Cardinality: 1 : M

3. LOAN — Loan_Payment — PAYMENT

- One loan can have many payments.
- Each payment belongs to one loan.
- Cardinality: 1 : M

4. CUSTOMER — Depositor — ACCOUNT

- One customer can have one or more accounts.
- Each account belongs to one customer.
- Cardinality: 1 : M

5. EMPLOYEE — Customer_Banker — CUSTOMER

- One employee can serve many customers.
- Each customer is assigned to one employee (banker).
- Cardinality: 1 : M

6. ACCOUNT — ISA — SAVINGS ACCOUNT

- An account can be classified as a Savings Account.

- A Savings Account inherits all attributes of the Account entity.
- Relationship: Specialization (ISA)

7. ACCOUNT — ISA — CHECKING ACCOUNT

- An account can be classified as a Checking Account.
- A Checking Account inherits all attributes of the Account entity.
- Relationship: Specialization (ISA)

3. Cardinality

Relationship	Cardinality
Customer — Borrower — Loan	1 : M
Branch — Loan_Branch — Loan	1 : M
Loan — Loan_Payment — Payment	1 : M
Customer — Depositor — Account	1 : M
Employee — Customer_Banker — Customer	1 : M
Account — ISA — Savings Account	Specialization
Account — ISA — Checking Account	Specialization

4. ER Diagram

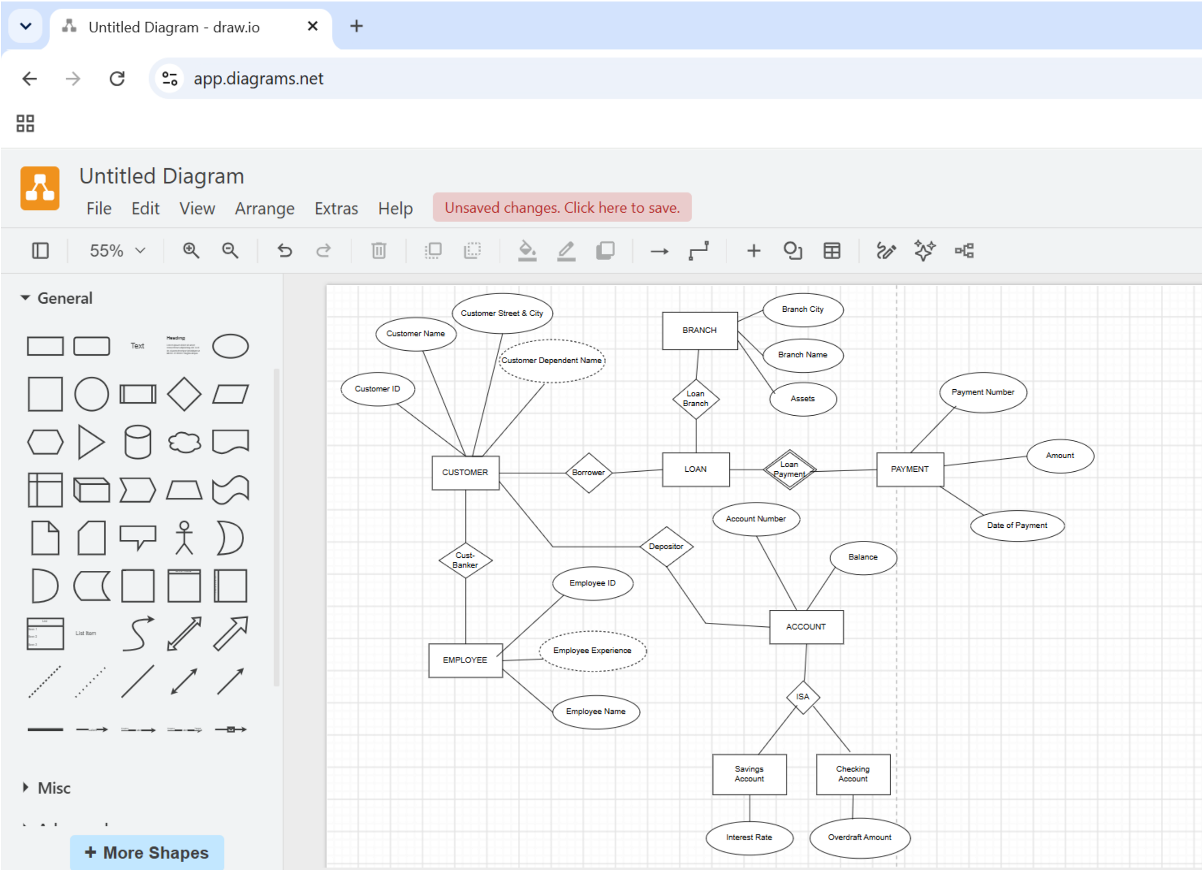

Fig: ER diagram showing Bank Management System

5. Explanation of the ER Diagram

The ER diagram represents a Bank Management System with the main entities Customer, Branch, Loan, Payment, Account, and Employee. A Customer can borrow a Loan through the Borrower relationship and can maintain an Account through the Depositor relationship. Each Loan is issued by a Branch and is associated with Payments through the Loan Payment relationship. The Customer–Banker relationship connects Customers with Employees. The Account entity is further specialized into Savings Account and Checking Account using the ISA relationship, where a Savings Account has an Interest Rate and a Checking Account has an Overdraft Amount. This ER diagram effectively models the entities, attributes, relationships, and specialization required for a Bank Management System.

Conclusion

The ER diagram of the Bank Management System identifies the major entities (Customer, Branch, Loan, Payment, Account, and Employee), their attributes, and the relationships among them. The use of ISA specialization for account types improves the database structure by reducing redundancy and supporting inheritance. This ER model provides a clear, efficient, and well-organized database design for banking operations.

3. Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.

Extended ER (EER) Model

The Extended Entity–Relationship (EER) model is an enhancement of the basic ER model. It provides additional concepts to represent complex real-world applications more accurately. EER supports inheritance, classification, and abstraction through generalization, specialization, and aggregation.

1. Generalization

Generalization is a bottom-up approach in which multiple lower-level entities with similar characteristics are combined into a higher-level entity.

Example:

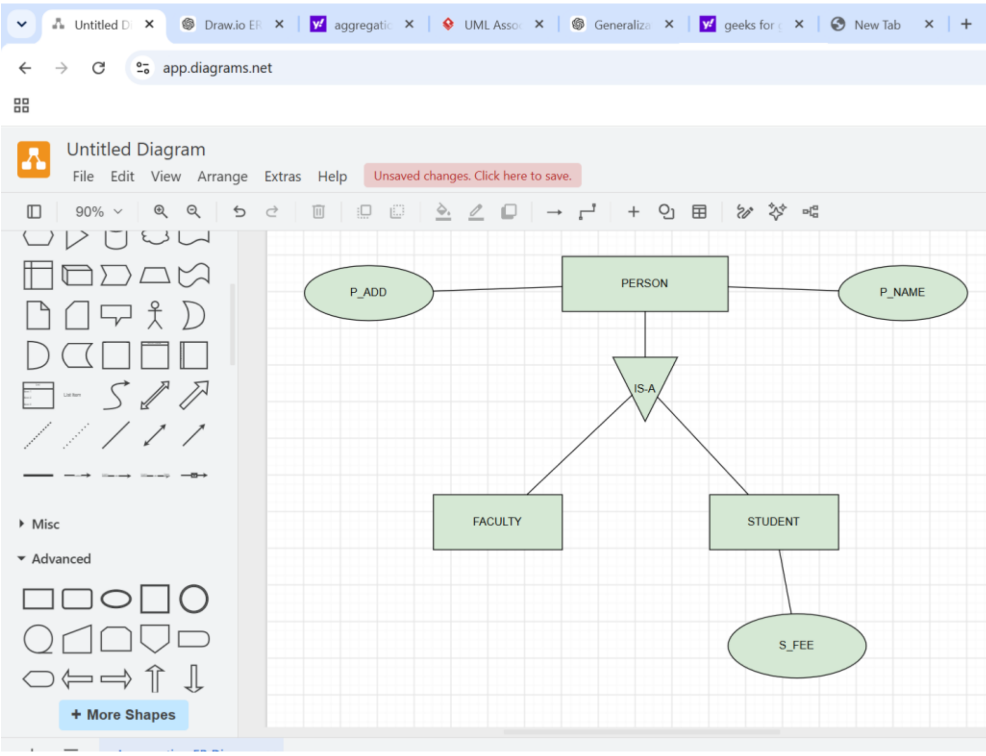

Fig: EER Diagram Representing Person Generalization

STUDENT and FACULTY can be generalized to a higher-level entity called PERSON as shown in diagram below. In this case, common attributes like P_NAME and P_ADD become part of a higher entity (PERSON) and specialized attributes like S_FEE become part of a specialized entity (STUDENT).

Purpose:

- Reduces redundancy.
- Represents common features at a higher level.
- Improves database organization.

2. Specialization

Specialization is a top-down approach in which a higher-level entity is divided into lower-level entities based on distinguishing characteristics.

Example:

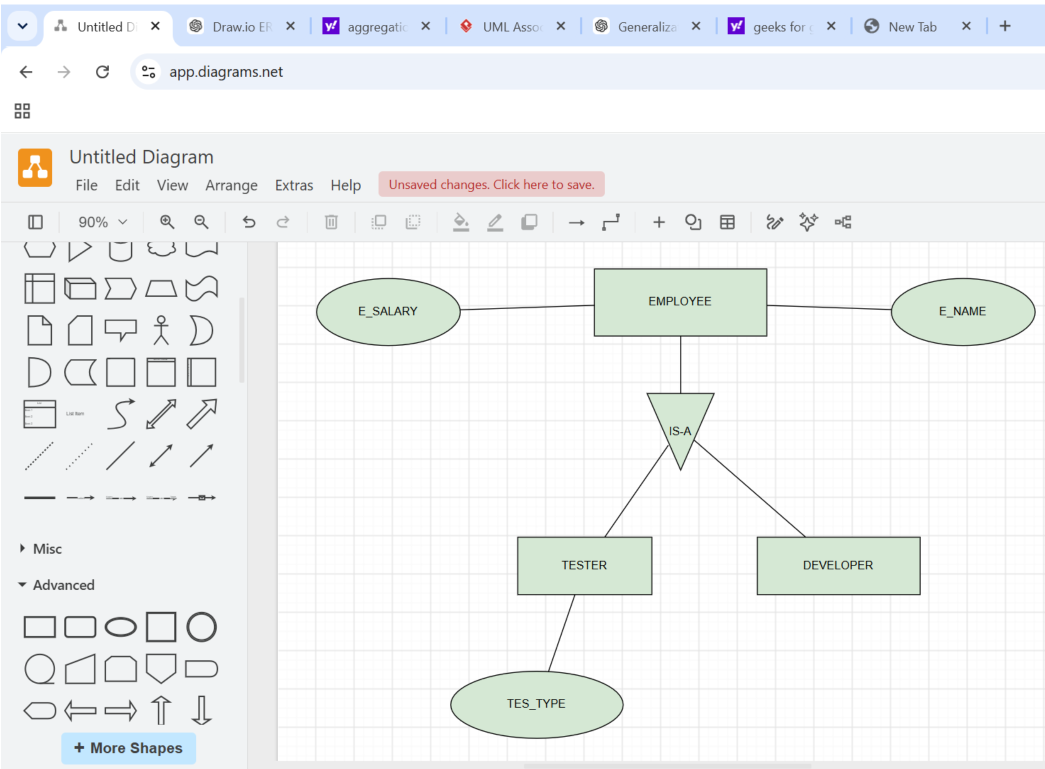

Fig: EER Diagram Representing Employee Specialization

an EMPLOYEE entity in an Employee management system can be specialized into DEVELOPER, TESTER, etc. In this case, common attributes like E_NAME, E_SAL, etc. become part of a higher entity (EMPLOYEE) and specialized attributes like TES_TYPE become part of a specialized entity (TESTER).

Purpose:

- Captures differences among entity subtypes.
- Allows subtype-specific attributes.
- Provides better modeling of real-world scenarios.

3. Aggregation

Aggregation is an abstraction mechanism that treats a relationship between entities as a higher-level entity.

Example:

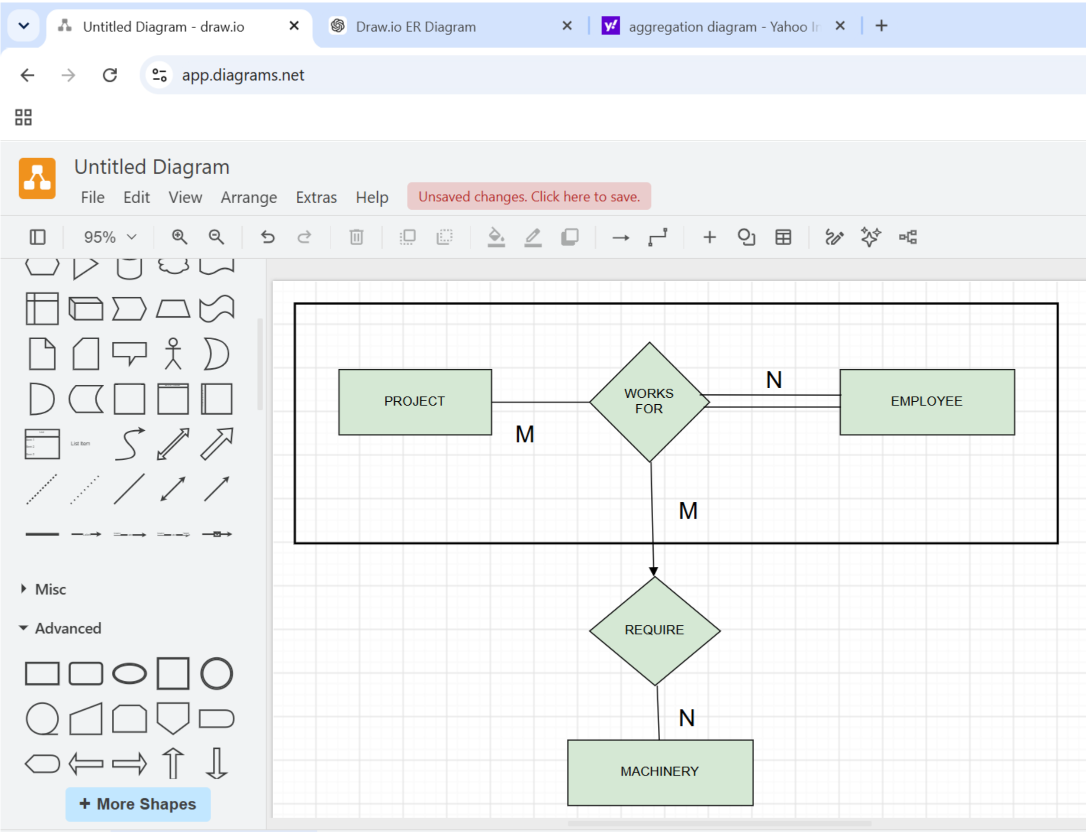

Fig: EER Diagram Illustrating Aggregation

an Employee working on a project may require some machinery. So, REQUIRE relationship is needed between the relationship WORKS_FOR and entity MACHINERY. Using aggregation, WORKS_FOR relationship with its entities EMPLOYEE and PROJECT is aggregated into a single entity and relationship REQUIRE is created between the aggregated entity and MACHINERY.

Purpose:

- Represents relationships among relationships.
- Simplifies complex ER diagrams.
- Provides higher-level abstraction.

Difference Between Generalization and Specialization

Generalization	Specialization
Bottom-up approach	Top-down approach
Combines entities	Divides entities
Creates a superclass	Creates subclasses
Reduces redundancy	Adds detailed distinctions

Conclusion

The EER model extends the ER model by supporting generalization, specialization, and aggregation. These concepts help represent inheritance, classification, and abstraction, making database design more accurate, scalable, and suitable for complex applications.

4. Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.

Hospital Management System – EER Design

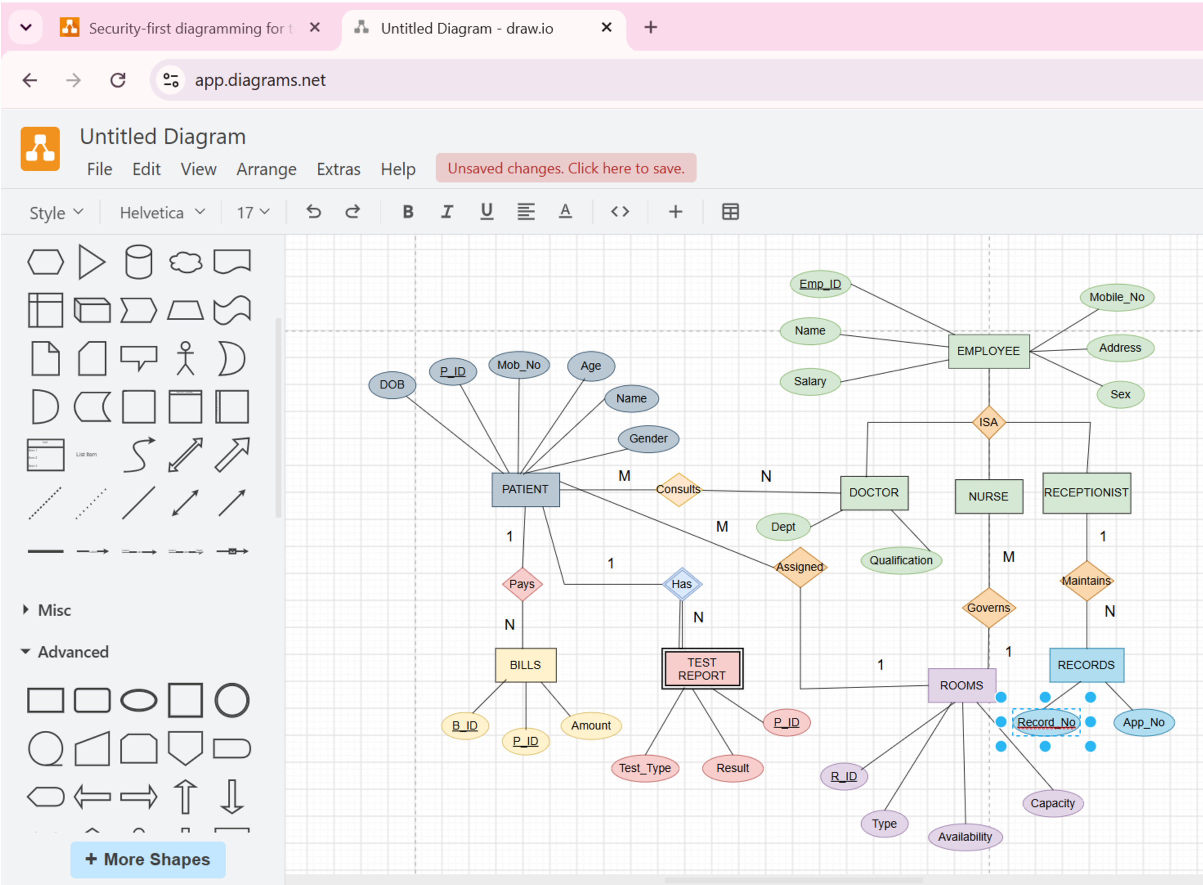

Fig: EER Diagram for Hospital Management System

This EER diagram illustrates the database structure of a Hospital Management System using entities, relationships, inheritance (ISA), and weak entities. It provides an efficient and scalable design for managing hospital operations and maintaining data integrity.

Main Entities

1. PATIENT (Strong Entity)

Attributes

- P_ID (Primary Key)

- Name
- Age
- Gender
- Mobile_No
- DOB

Responsibilities

- Consults doctors
- Pays bills
- Undergoes medical tests
- Receives treatment

2. EMPLOYEE (Supertype)

Attributes

- Emp_ID (Primary Key)
- Name
- Salary
- Address
- Mobile_No
- Sex

The Employee entity represents all hospital staff and acts as the supertype.

3. DOCTOR (Subtype)

Inheritance from Employee

Additional Attributes

- Department
- Qualification

Responsibilities

- Consults patients
- Assigned to rooms

4. NURSE (Subtype)

Inheritance from Employee

Responsibilities

- Governs patient rooms
- Assists doctors
- Monitors patients

5. RECEPTIONIST (Subtype)

Inheritance from Employee

Responsibilities

- Maintains patient records
- Manages appointments
- Registers patients

6. ROOMS

Attributes

- R_ID (Primary Key)
- Type
- Capacity
- Availability

Used for patient admission and treatment.

7. BILLS

Attributes

- B_ID (Primary Key)
- Amount

Represents payment details generated for patients.

Weak Entity

TEST_REPORT (Weak Entity)

Weak Entity because

A Test Report cannot exist without a Patient.

It depends on the Patient for identification.

Attributes

- Test_Type
- Result

Partial Key

- Report_Number (if included)

Identifying Relationship

- Patient Has Test Report

This satisfies the weak entity concept in EER modeling.

Inheritance (Specialization)

- Employee is the supertype.
- Doctor, Nurse, and Receptionist are the subtypes.
- These subtypes inherit the common attributes of Employee and perform their own specialized roles in the hospital.

Inherited Attributes:

- Emp_ID
- Name
- Address
- Mobile_No
- Salary

Specialized Responsibilities:

- Doctor: Diagnoses and treats patients.
- Nurse: Assists doctors and cares for patients.
- Receptionist: Manages appointments and patient records.

Conclusion

The Hospital Management System EER diagram efficiently models hospital operations using strong entities, weak entities, inheritance (ISA), attributes, and relationships. The Employee supertype with specialized Doctor, Nurse, and Receptionist subtypes demonstrates inheritance, while Test_Report acts as a weak entity dependent on Patient. This design improves data integrity, minimizes redundancy, and provides a scalable database structure suitable for modern healthcare systems.

5. Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.

Role of a DBA

A Database Administrator (DBA) is responsible for managing, maintaining, securing, and optimizing the database system of an organization. The DBA ensures that the database is available, reliable, secure, and performs efficiently.

Responsibilities of a DBA

1. Database Installation and Configuration

- Install DBMS software such as MySQL, Oracle Database, or Microsoft SQL Server.
- Configure database parameters and storage settings.

2. Database Design and Maintenance

- Create and maintain database schemas.
- Define tables, constraints, indexes, and relationships.
- Modify structures when business requirements change.

3. Security Management

- Create user accounts and assign privileges.
- Prevent unauthorized access.
- Implement authentication and authorization policies.

4. Backup and Recovery

- Perform regular database backups.
- Restore data after failures or data loss.
- Develop disaster recovery plans.

5. Performance Monitoring and Tuning

- Monitor query performance.

- Optimize indexes and SQL statements.
- Manage memory, storage, and system resources.

6. Data Integrity and Consistency

- Enforce constraints and validation rules.
- Ensure transactions maintain database consistency.
- Prevent duplicate or invalid data.

7. Concurrency and Transaction Management

- Manage multiple users accessing the database simultaneously.
- Prevent conflicts using locking and transaction control.

8. Capacity Planning

- Monitor database growth.
- Plan future storage and performance requirements.

Importance of a DBA

A DBA plays a critical role in ensuring:

- High availability of the database.
- Data security and confidentiality.
- Efficient performance for applications.
- Reliable backup and recovery mechanisms.
- Data integrity and consistency across the organization.

Conclusion

The Database Administrator (DBA) is responsible for the complete lifecycle management of a database system. By handling installation, security, backup, recovery, performance tuning, and maintenance, the DBA ensures that organizational data remains secure, accurate, available, and efficient for all users and applications.